

Delta Robot with Damped Articulations

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Technical summary of the 2017 undergraduate thesis

Abstract—This project develops a Delta robot prototype and models its coupled geometry, compliant subsystems, and trajectory control. Nonlinear constraints link joint motion, platform position, and internal deformation. A rigid approximation provides explicit kinematics; an extended constrained framework introduces spring–damper states for further dynamic analysis.

Index Terms—Delta robot, constrained dynamics, compliance, state estimation.

I. COUPLED GEOMETRIC MODEL

Three kinematic chains connect actuated upper arms to a common moving platform through paired lower members [1]. The shared platform makes the chains interdependent through nonlinear geometric constraints. The nominal rigid model has three translational degrees of freedom and assumes symmetric chains, a fixed reference platform, and zero damper compression.

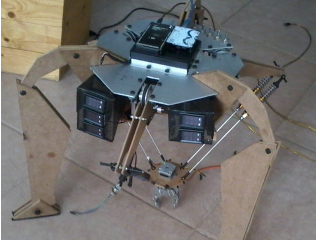


Fig. 1. Constructed Delta prototype (left) and CAD model (right).

Forward kinematics locates the platform through sphere intersection; inverse kinematics recovers compatible joint angles from a Cartesian target. Multiple solutions require geometric branch selection. With coordinates collected in $\mathbf{q}$, closure constraints Φ and their Jacobian $\mathbf{J}_\Phi$ satisfy

$$\Phi(\mathbf{q}) = \mathbf{0}, \quad \mathbf{J}_\Phi(\mathbf{q})\dot{\mathbf{q}} = \mathbf{0}. \quad (1)$$

Differentiation yields configuration-dependent velocity and acceleration mappings specific to the rigid approximation.

II. COMPLIANCE AND INTERNAL STATES

The compliant model augments joint coordinates with spring compression and its time derivative. Lower-member compliance is intended to soften deceleration loads and isolate end-effector vibration; compliant supports also maintain transmission tension. Elastic and viscous forces depend on deformation and its rate, respectively.

The extended representation includes attachment-point geometry and platform-frame position and orientation. These

variables must satisfy linkage constraints throughout deformation, relating internal motion, elastic energy, dissipation, and changing force directions to the same configuration.

III. DYNAMIC FORMULATION AND DESIGN SEARCH

Newtonian joint balances relate actuator input, transmitted torque, inertia, and gravity under simplifying load assumptions. Appendix D outlines a constrained Lagrangian framework for rigid and compliant representations, combining energy descriptions with geometric constraints and reaction forces. The explicit control development uses a reduced dynamic model.

Appendix C adds a geometric design problem: an evolutionary search varies link geometry and evaluates approximate reachable workspace volume, connecting model parameters to a design objective through repeated kinematic evaluation.

IV. FROM THE MODEL TO CONTROL AND ESTIMATION

Desired Cartesian trajectories are mapped into joint references through inverse kinematics. Local linearization describes dynamics around the reference motion; continuous-to-discrete transformations support a sampled implementation. The analytical actuator model treats current as the control input under an assumed inner current-control loop.

Linear-quadratic tracking balances reference error against control effort. Backward integration of a Riccati equation and a reference-dependent term determines the feedback and tracking law. A Luenberger observer reconstructs angular velocity from position measurements and known control inputs, with estimation-error dynamics used to choose observer gains. Precomputed matrices and interpolation reduce the proposed online computation.

V. EVIDENCE AND MODEL VALIDATION

The thesis documents prototype construction and presents analytical tracking responses to smooth and discontinuous references, excluding the observer. Experimental closed-loop performance remains unverified. The compliant framework requires completion, parameter identification, and validation against measured deformation and motion to assess the model's predictive accuracy.

REFERENCES

- [1] S. Restrepo Ruiz, “Construcción de un prototipo de robot Delta con articulaciones amortiguadas,” Undergraduate thesis, Electronic Engineering, Universidad del Quindío, Armenia, Colombia, Nov. 2017. [Online]. Available: https://github.com/savethebeesandseeds/robot_delta_doc